

Business Requirements Document

Enterprise Learning Management System (LMS)

Multi-tenant training platform · Microservice architecture

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1. Introduction

1.1 Purpose

This Business Requirements Document (BRD) defines the business, functional, and non-functional requirements for the Enterprise Learning Management System (LMS) — a multi-tenant training platform. It describes the scope, stakeholders, requirements, microservice solution architecture, key process flows, data model, integration and security requirements, and acceptance criteria, in order to align stakeholders and guide the design and delivery of the platform.

1.2 Intended audience

Product owners, solution and enterprise architects, backend and front-end engineers, quality assurance, DevOps and platform operations, information security, and compliance stakeholders.

1.3 Document conventions

Shall / must denotes a mandatory requirement; **should** denotes a recommended requirement; **may / could** denotes an optional capability. Requirement identifiers follow the pattern FR-`<module>-<n>` and NFR-`<n>`.

1.4 Revision history

Version	Date	Author	Description
1.0	21 June 2026	Gs Dwivedi	Initial enterprise draft for stakeholder review.

2. Executive Summary

The Enterprise LMS is a multi-tenant training platform serving universities, colleges, and organisations across both university and corporate training segments. It enables administrators, managers, trainers, organisers, and students to manage the complete learning lifecycle — identity and access, organisations, courses and content, batches and enrolment, assessments, video-based delivery, engagement, and multi-channel notifications — on a single, secure, and extensible platform.

A defining capability is a fully dynamic role-based access model: roles, fine-grained permissions, and the modules each role can see are stored as configuration and can be reconfigured at runtime without code changes or redeployment. The platform is built as a set of independent microservices on a Spring Boot backend with a Next.js front end, communicating asynchronously over an event bus, which enables independent scaling, resilience, and rapid evolution of individual capabilities.

3. Business Context

Training delivery currently spans multiple disconnected tools, with inconsistent access control, fragmented content and assessment management, and ad-hoc communication. There is a clear need for a unified platform that can serve multiple institutions and organisations from a single deployment, enforce precise and adaptable access control, deliver structured learning content and assessments, stream video lessons securely, and communicate reliably across email, WhatsApp, and SMS — including the automated generation and delivery of official documents.

The platform must support a layered organisational hierarchy (administrators, managers, universities and organisations, trainers and organisers, and students), enforce governance over sensitive operations, and remain extensible so that new roles, modules, channels, and integrations can be added with minimal disruption.

4. Scope

4.1 In scope

- A multi-tenant LMS serving universities, colleges, and organisations from a single platform.
- A fully dynamic role-based access control model with runtime-configurable roles, permissions, and module visibility.
- Organisation and tenancy management, including trainer-to-subject affiliations.
- User management and provisioning with governance and approval workflows.
- Course and content management (modules, submodules, notes, PDF, PPT, comparison tables, video).
- Batch and enrolment management with scheduling.
- Assessment and evaluation (theoretical, practical, and theory-based) with online marking.
- Secure video delivery via adaptive, chunked streaming.
- Engagement and feedback (comments, threaded replies, trainer feedback).
- Multi-channel notification and document delivery (email, WhatsApp, SMS) with JasperReports document generation.
- Administration, configuration, audit, and reporting.
- A microservice solution architecture on Spring Boot with a Next.js front end.

4.2 Out of scope

- Procurement and contractual setup of third-party provider accounts (email, WhatsApp, SMS, SSO).
- Native mobile applications beyond the responsive web and mobile experience (candidate for a later phase).
- Marketing / bulk-campaign management.
- Finance, payroll, and HR systems outside the training domain.

5. Business Objectives & Success Criteria

Objective	Success criterion
Unify training delivery on one platform	Universities, organisations, trainers, and students operate from a single multi-tenant deployment.
Make access control fully adaptable	Roles, permissions, and module visibility can be changed at runtime, taking effect within seconds and without redeployment.
Automate official document delivery	Onboarding and reporting documents are generated and delivered automatically with tracked status.
Ensure reliable, secure delivery	Video and documents are delivered securely; notifications are at-least-once with retry and audit.
Scale and evolve independently	Each capability scales independently; new roles, modules, and channels are added by configuration.

6. Stakeholders & User Roles

6.1 Stakeholders

Stakeholder	Interest
Platform administrators	Configure the platform, roles, and governance; oversee operations and audit.
Managers	Administer assigned universities, trainers, and operations within their scope.
University & organisation officials	Receive official documents and oversee their institution's presence.
Trainers & organisers	Create and deliver content, manage batches and assessments, and communicate with learners.
Students	Access assigned learning, take assessments, watch video lessons, and receive notifications.
Platform operations	Operate, monitor, scale, and secure the platform.
Information security & compliance	Ensure access control, data protection, consent, and retention obligations are met.

6.2 User roles

The platform ships with the following baseline roles. Because the access model is dynamic, administrators may create additional roles (for example, a Counsellor role) and adjust the access of any role at runtime.

Role	Summary of responsibilities
Admin	Highest authority; configures the platform, manages all entities and roles, and approves governed actions.
Manager	Created only by an Admin; administers assigned universities, colleges, and trainers; may deactivate entities; destructive deletes require Admin approval.
Trainer	Adds students, creates batches, authors courses and content, sets assessments, and sends notifications, scoped to affiliated universities and subjects.
Organiser	Performs trainer-equivalent delivery and organisation tasks, including course authoring and assessment.
Student	Accesses assigned courses, takes assessments, watches video lessons, comments, and submits feedback.

7. Functional Requirements

7.1 Identity, Access & Roles

ID	Requirement	Priority	Description
FR-IAM-01	Authentication	Must	The platform shall authenticate users and issue a session token; SSO / OAuth integration shall be supported for institutional sign-in.
FR-IAM-02	Dynamic roles	Must	Roles shall be data-driven; administrators shall create, edit, and remove roles at runtime (e.g., a Counsellor role) without code changes.
FR-IAM-03	Fine-grained permissions	Must	Access shall be governed by fine-grained permission codes mapped to roles.
FR-IAM-04	Dynamic module visibility	Must	The set of UI modules a role can see shall be configurable at runtime; the application menu shall render from this configuration.
FR-IAM-05	Role assignment & scoping	Must	Users shall receive access only through role assignment; assignments shall support a tenant scope (e.g., Manager of University X).
FR-IAM-06	Per-user overrides	Should	Administrators may grant or deny individual permissions to a specific user as an exception.
FR-IAM-07	Instant propagation	Must	Changes to a role's permissions or modules shall take effect for affected users within seconds, without requiring re-login.
FR-IAM-08	System-role protection	Must	Built-in roles (e.g., Admin) shall be protected from deletion and from changes that would remove the ability to administer the platform (lockout prevention).

7.2 Organisation & Tenancy

ID	Requirement	Priority	Description
FR-ORG-01	Manage organisations	Must	The platform shall manage universities, colleges, and organisations.
FR-ORG-02	Multi-tenancy	Must	Tenant data shall be isolated using a shared schema with a tenant discriminator; cross-tenant access shall be prevented.
FR-ORG-03	Trainer-subject affiliation	Must	Trainers shall be affiliated to universities and subjects in a many-to-many relationship (a university may have many trainers; a subject may have many trainers).
FR-ORG-04	Official contact	Must	Each university or organisation shall capture an official contact for communications.

7.3 User Management & Provisioning

ID	Requirement	Priority	Description
FR-USR-01	Create users & assign roles	Must	Administrators and authorised roles shall create users; access shall flow from the assigned role, never copied onto the user.
FR-USR-02	Trainer provisioning	Must	Trainer creation shall provision identity and role and the organisational affiliation consistently across services, with compensation on failure.
FR-USR-03	Student provisioning	Must	Students shall be created and assigned the appropriate role and enrolment.
FR-USR-04	Activate / deactivate	Must	Users and entities shall be deactivatable (reversible) and reactivatable.
FR-USR-05	Restricted creation	Must	Creation of a Manager shall be restricted to an Admin.

7.4 Governance & Approval

ID	Requirement	Priority	Description
FR-GOV-01	Approval-gated deletes	Must	Destructive actions by constrained roles (e.g., a Manager delete) shall require approval per a configurable policy; the resource shall not be removed until approved.
FR-	Approve / reject	Must	Pending requests shall be visible to approvers;

ID	Requirement	Priority	Description
GOV-02			approval executes the action and rejection discards it and restores the resource.
FR-GOV-03	Deactivate vs delete	Must	Deactivation shall be immediate and reversible; deletion shall be governed.
FR-GOV-04	Configurable policy	Should	The mapping of role and action to approval requirement shall be configurable at runtime.

7.5 Course & Content Management

ID	Requirement	Priority	Description
FR-CRS-01	Course structure	Must	The platform shall support courses with modules and submodules in tutorial order.
FR-CRS-02	Content items	Must	Content shall include notes, PDF, PPT, comparison tables, and video.
FR-CRS-03	Authoring	Must	Trainers and organisers with the relevant permissions shall author content, with drag-and-drop ordering.
FR-CRS-04	Course assignment	Must	Courses shall be assignable to batches and students; students shall gain read access only when a course is assigned to them.

7.6 Batch & Enrolment

ID	Requirement	Priority	Description
FR-BAT-01	Manage batches	Must	Trainers and organisers shall create batches and add students.
FR-BAT-02	Schedule	Must	Each batch shall have a schedule (batch time) visible to enrolled students.
FR-BAT-03	University scope	Must	Batches shall be scoped to a university.
FR-BAT-04	Enrolment	Must	The platform shall manage enrolment of students into batches and courses.

7.7 Assessment & Evaluation

ID	Requirement	Priority	Description
FR-ASM-01	Assessment types	Must	The platform shall support theoretical, practical, and theory-based assessments.
FR-ASM-02	Assignments & tests	Must	Assignments and tests shall be created and tied to a batch.
FR-ASM-03	Online evaluation	Must	Students shall submit online and submissions shall be evaluated to produce marks.
FR-ASM-04	Results	Must	Marks and results shall be surfaced to the relevant students.

7.8 Learning Delivery & Video

ID	Requirement	Priority	Description
FR-VID-01	Adaptive streaming	Must	Video lessons shall be delivered via adaptive, chunked streaming (HLS).
FR-VID-02	No full download	Must	Videos shall not be downloadable as a single file; access shall be via short-lived signed URLs bound to the student's enrolment.
FR-VID-03	Progress	Should	The platform shall track each student's viewing progress.

7.9 Engagement & Feedback

ID	Requirement	Priority	Description
FR-ENG-01	Comments & replies	Must	Students shall comment on courses and videos, with threaded replies.
FR-ENG-02	Trainer notification	Must	The relevant trainer shall be notified of a comment and able to reply; the student shall be notified of the reply.
FR-ENG-03	Feedback routing	Must	Students shall submit feedback on trainers, routed to the relevant manager.

7.10 Notification & Document Delivery

ID	Requirement	Priority	Description
FR-NOT-01	Multi-channel	Must	Event-driven notifications shall be delivered over email and WhatsApp, with an extensible adapter model for further channels such as SMS.
FR-NOT-02	Document generation	Must	Documents shall be generated via JasperReports (PDF / XLSX / DOCX) and attached to messages.
FR-NOT-03	Onboarding documents	Must	Creating a university shall generate the official onboarding document set and deliver it to the official contact.
FR-NOT-04	Reliable delivery	Must	Delivery shall be tracked, retried with back-off, dead-lettered on exhaustion, and idempotent against duplicate events.

7.11 Administration & Configuration

ID	Requirement	Priority	Description
FR-ADM-01	Access configuration UI	Must	An administration interface shall provide role-permission and role-module matrices to configure access.
FR-ADM-02	Site settings	Should	Administrators shall manage platform and tenant settings.
FR-ADM-03	Dashboards	Should	Operational dashboards shall present delivery, governance, and activity status.

7.12 Audit & Reporting

ID	Requirement	Priority	Description
FR-AUD-01	Audit trail	Must	Sensitive actions (access changes, approvals, provisioning, deletions) shall be recorded in an audit trail.
FR-AUD-02	Reporting	Should	The platform shall provide reporting on activity and delivery.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Read-heavy workloads shall be optimised with caching and read replicas; video shall be served via CDN; standard document generation shall complete within the agreed SLA (target ≤ 10 s, 95th percentile).
NFR-02	Scalability	Services shall be stateless behind the gateway and scale horizontally; report generation, media, and channel workers shall scale independently.
NFR-03	Availability & reliability	Target availability $\geq 99.5\%$; inter-service calls shall use resilience patterns (timeout, retry with back-off, circuit breaker, bulkhead); delivery shall be at-least-once with idempotent consumers.
NFR-04	Security	All access shall be authenticated and authorised via the dynamic RBAC model; transport shall use TLS; secrets shall be held in a secrets manager; media and documents shall use short-lived signed URLs.
NFR-05	Compliance	Tenant data shall be isolated; messaging shall respect consent and regional regulations; data retention shall follow policy.
NFR-06	Observability	The platform shall emit metrics, structured logs, and distributed traces, and alert on error-rate, latency, and dead-letter thresholds.
NFR-07	Maintainability & extensibility	Services shall have clean module boundaries; new roles, modules, channels, and event types shall be addable by configuration and pluggable adapters.
NFR-08	Usability	The web and mobile experience shall be responsive and present role-appropriate navigation.
NFR-09	Data integrity & consistency	Each service shall own its data (database per service); cross-service consistency shall be achieved via events and the saga pattern with compensation.

9. Solution Architecture

9.1 System context

The Enterprise LMS is accessed by five baseline user roles through a single web and mobile front end, and integrates with external systems for identity, document generation, messaging, and media delivery.

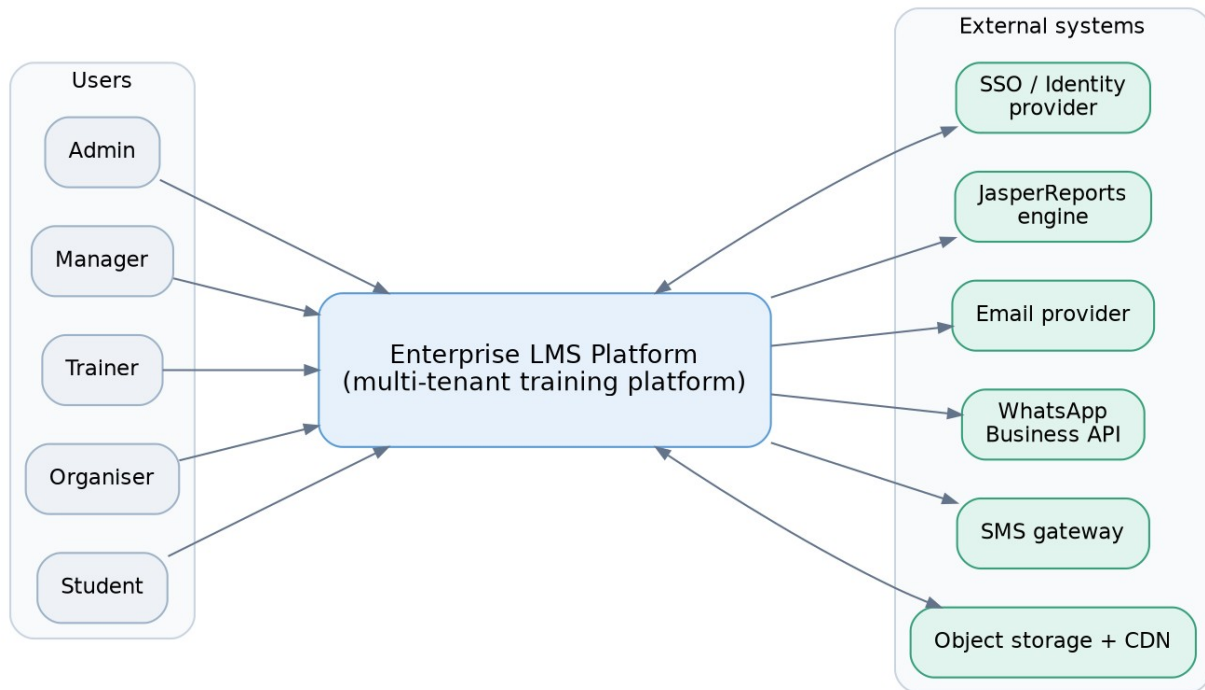


Figure 1 — System context.

9.2 Microservice architecture

The platform follows a microservice architecture. A Next.js front end calls a Spring Cloud Gateway, which validates the session token and routes to independent Spring Boot services. Services communicate asynchronously over an Apache Kafka event bus, each owns its own data store, and shared platform services provide discovery and centralised configuration.

Resilience patterns protect inter-service calls.

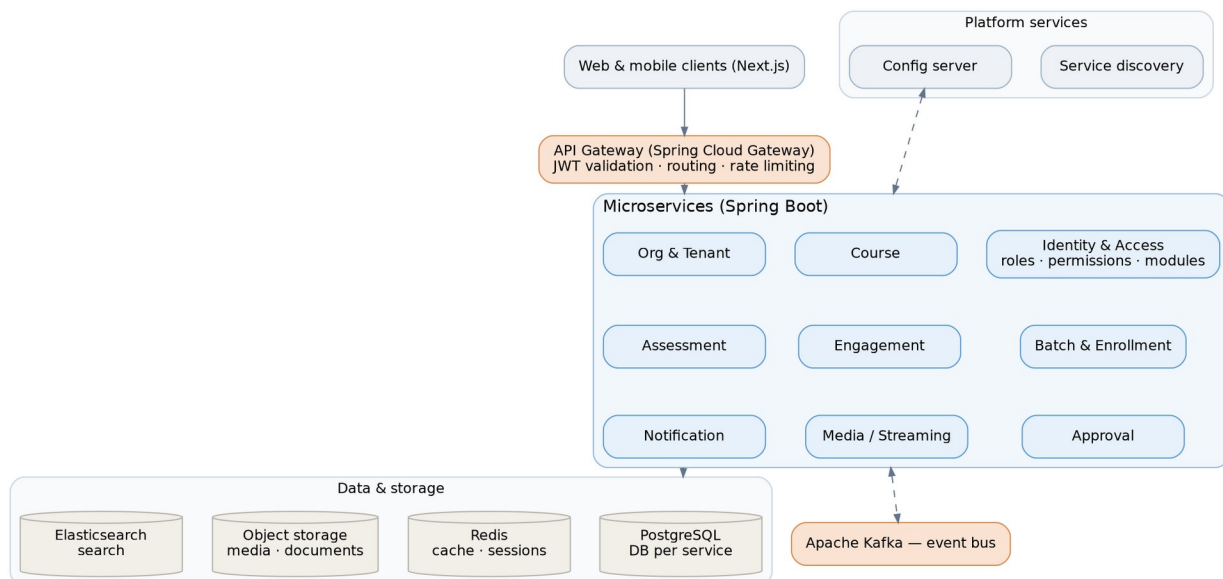


Figure 2 — Microservice architecture.

9.3 Service catalogue

Service	Responsibility
API Gateway	Single entry point; validates the token, routes requests, and applies rate limiting.
Identity & Access (IAM)	Authentication and the dynamic authorisation model: roles, permissions, modules, mappings, and scoping.
Org & Tenant	Universities, colleges, organisations, trainer-subject affiliations, and tenancy.
Course	Courses, modules, submodules, content items, and course assignment.
Batch & Enrolment	Batches, schedules, and enrolment of students.
Assessment	Assessments, assignments, tests, online evaluation, and marks.
Engagement	Comments, threaded replies, and feedback.
Approval	Pending governed actions and the configurable approval policy.
Notification	Multi-channel delivery and JasperReports document generation.
Media / Streaming	Video transcoding, adaptive streaming, signed URLs, and file storage.

9.4 Architecture principles

- Each service owns its own database; services never read another service's tables directly.
- Cross-service communication is asynchronous and event-driven over Kafka; synchronous calls are reserved for request-time needs.
- Services are stateless behind the gateway and scale horizontally and independently.
- The session token carries identity and roles only; permissions are resolved per service and cached, so access changes propagate by cache invalidation rather than re-login.
- Inter-service calls are protected by timeout, retry with back-off, circuit breaker, and bulkhead patterns.
- The boundaries are designed so the platform can also run as a modular monolith and be split under load without changing the model or APIs.

10. Key Process Flows

10.1 Request authorisation and dynamic access

Every action is authorised at request time, and access changes take effect without redeployment or re-login. The gateway validates the token; the target service resolves the user's effective permissions (from their roles, cached) and checks the required permission for the action and scope. When an administrator changes a role's permissions or modules, the cached permissions of affected users are invalidated so the next request reflects the new rules.

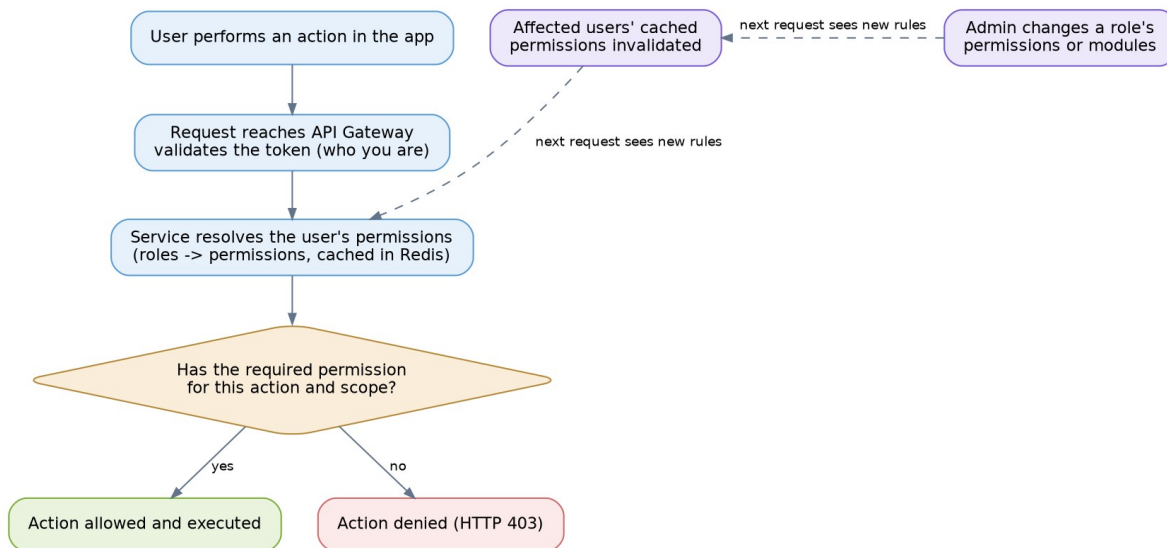


Figure 3 — Request authorisation and dynamic access propagation.

10.2 Governance — approval workflow

Destructive actions by constrained roles are governed. When a Manager requests a delete, the service checks whether the role and action require approval. If so, the action is held as a pending request and the resource is left intact until an Admin approves (executing the delete) or rejects (restoring the resource).

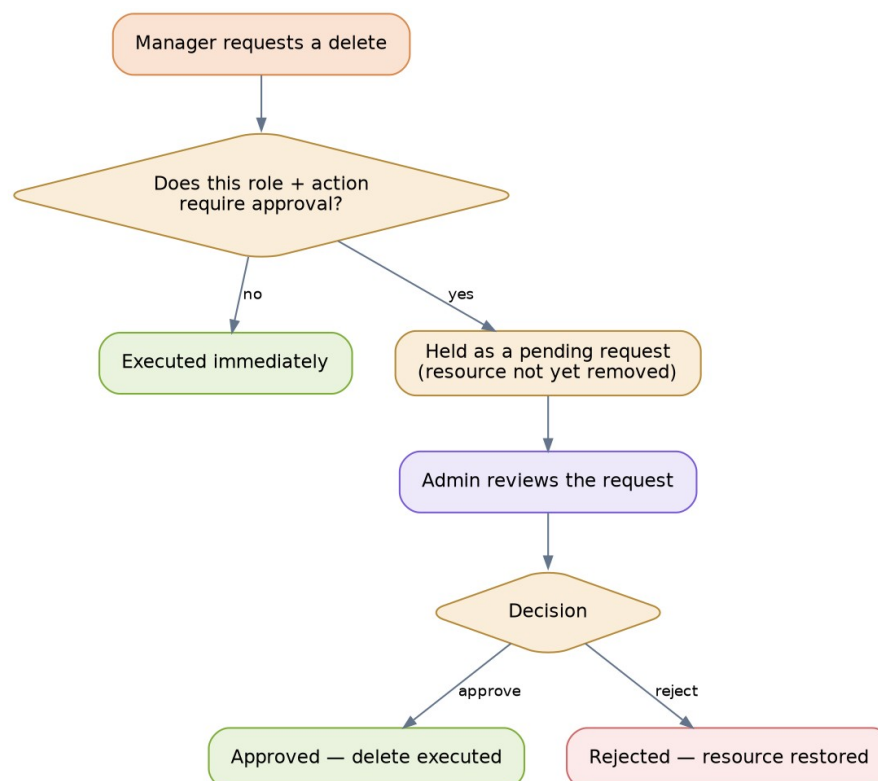


Figure 4 — Governance and approval workflow.

11. Data Model

Key entities, grouped by owning service, are summarised below. Each service owns and persists its own entities.

Entity (service)	Key attributes	Description
user (IAM)	id, email, status, perm_version	A platform user and authentication principal.
role / permission / module (IAM)	id, name/code, is_system, parent_id	The dynamic building blocks of access.
user_role / role_permission / role_module (IAM)	user_id, role_id, scope_id; role_id, permission_id; role_id, module_id	Mappings that grant roles to users and permissions and modules to roles.
university / organisation (Org)	id, name, type, official_contact	A tenant institution or organisation.
trainer_assignment (Org)	id, trainer_id, university_id, subject_id	A trainer's affiliation to a university and subject.
course / module / content_item (Course)	id, parent_id, position, type, storage_ref	Course structure and content.
batch / enrolment (Batch)	id, university_id, schedule; batch_id, student_id	Student groups, schedules, and enrolment.
assessment / submission (Assessment)	id, type, batch_id; assessment_id, student_id, marks	Assessments and evaluated submissions.
approval_request (Approval)	id, requested_by, action, resource_ref, status	A governed action awaiting decision.
notification / delivery (Notification)	id, event_type, status; channel, status, attempts	A notification and its per-channel delivery.

12. Integration Requirements

Integration	Type	Notes
SSO / Identity provider	OAuth / SAML	Institutional single sign-on for universities and organisations.
JasperReports	Embedded engine / report server	Generates PDF / XLSX / DOCX documents from report designs.
Apache Kafka	Event streaming	Asynchronous events between services; dead-letter topics for poison messages.

Integration	Type	Notes
Email provider (SES / SendGrid)	SMTP / API	Transactional email with attachments and delivery webhooks.
WhatsApp Business API	REST API	Template and document messages, subject to consent and session rules.
SMS gateway	REST / SMPP	Text notifications with regional routing compliance.
Object storage + CDN (S3 / MinIO)	S3 API	Stores media and documents; serves video via CDN with signed URLs.

13. Security & Compliance

- Authentication for all access, with SSO for institutions; the gateway validates every request.
- Authorisation through the dynamic RBAC model, enforced at the gateway and by method-level checks in each service.
- Tenant isolation so that data of one institution is never accessible to another.
- Short-lived signed URLs for video and documents, bound to the recipient, with no sensitive data in URLs.
- Secrets and provider credentials held in a secrets manager; TLS for all transport.
- An audit trail of access changes, approvals, provisioning, and deletions.
- Consent and opt-out for WhatsApp and SMS, and adherence to data-retention and regional data-protection requirements.

14. Assumptions, Dependencies & Constraints

14.1 Assumptions

- Event payloads carry the data required, or the relevant source-service APIs are available for enrichment.
- Official contacts for universities and organisations are captured at creation.
- Provider templates (WhatsApp) and routes (SMS) are approved before go-live.

14.2 Dependencies

- Apache Kafka, object storage and CDN, and managed databases are provisioned.
- Provider accounts (SSO, email, WhatsApp, SMS) with valid credentials.
- JasperReports designs are authored and version-controlled.

14.3 Constraints

- WhatsApp messaging is bound by template-approval and session-window rules.
- Provider rate limits and attachment-size limits apply.
- Content and documents must comply with branding and legal requirements.

15. Acceptance Criteria

- Access is governed entirely by the dynamic RBAC model; roles, permissions, and modules can be changed at runtime and take effect within seconds without re-login.
- Users receive access only through role assignment; creating a user with a given role grants exactly that role's access.
- Destructive actions by constrained roles are correctly gated by the approval workflow, and lockout-prevention safeguards hold.
- Courses, batches, assessments, and content behave per the functional requirements, scoped correctly per tenant.
- Video is delivered by adaptive streaming and cannot be downloaded as a single file.
- Notifications and official documents are generated and delivered across channels, with tracked status, retry, and audit.
- Each service owns its data, and cross-service operations remain consistent through events and compensation.

16. Risks & Mitigations

Risk	Likelihood	Mitigation
Distributed-transaction inconsistency across services	Medium	Saga pattern with compensation, idempotency keys, and the transactional outbox where stronger guarantees are needed.
Misconfiguration of dynamic access (lockout)	Medium	System-role protection, last-admin and governing-permission safeguards, and a protected break-glass account.
Provider outage or rate limiting	Medium	Retry with back-off, dead-letter, optional multi-provider fallback, and alerting.
Video piracy or link sharing	Medium	Chunked streaming, short-lived signed URLs bound to enrolment, and no direct storage exposure.
Load spikes on assessment or reporting	Medium	Independent scaling, queue-based load levelling, and caching.
Tenant data leakage	Low–Med	Tenant isolation, scoped permissions, and access auditing.

17. Glossary

Term	Definition
RBAC	Role-Based Access Control — access granted via roles and permissions.
Multi-tenancy	Serving multiple isolated institutions or organisations from a single

Term	Definition
	deployment.
Microservice	An independently deployable service that owns a single business capability and its data.
Saga	A pattern for cross-service consistency using a sequence of local steps with compensating actions.
Idempotency	Processing the same event more than once has the same effect as once.
HLS	HTTP Live Streaming — adaptive, chunked video delivery.
JasperReports / .jrxml	A reporting engine and its design format used to generate PDF / XLSX / DOCX.
DLQ	Dead-letter queue for messages that repeatedly fail.
Signed URL	A time-limited, access-controlled link to a stored object.